

Lec 01 - Set Theory

Set: well-defined collection of objects. Element: object in a set. Sets usually uppercase A,B; elements lowercase x,y. $x \in A$ means x is in A; $x \notin A$ means x is not in A. $\emptyset$ or $\{\}$ = empty set. U or Ω = universal set. $|A|$ = cardinality.

Forms: listing $A = \{1,2,3\}$; descriptive; set-builder $A = \{x \mid 1 \leq x \leq 6, x \in \mathbb{Z}\}$.

Subset $A \subseteq B$	Every element of A is in B
Proper $A \subset B$	$A \subseteq B$ and $A \neq B$
$P(A)$	Power set: all subsets of A
$P(A)$	If $ A =n$, then $ P(A) =2^n$
Intersection	$A \cap B$: common elements
Union	$A \cup B$: in A or B or both
Complement	A^c or $\bar{A}$: in U but not A
Difference	$A - B$: in A not B = $A \cap B^c$
Disjoint	$A \cap B = \emptyset$
Cartesian $A \times B$	$\{(a,b) \mid a \in A, b \in B\}$; $ A \times B = A B $

Cardinality

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Set identities

Commutative: $A \cap B = B \cap A$; $A \cup B = B \cup A$. Associative: $(A \cap B) \cap C = A \cap (B \cap C)$; $(A \cup B) \cup C = A \cup (B \cup C)$. Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$; $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Identity: $A \cap U = A$; $A \cup \emptyset = A$. Domination: $A \cup U = U$; $A \cap \emptyset = \emptyset$. Idempotent: $A \cap A = A$; $A \cup A = A$. Complement: $A \cap A^c = \emptyset$; $A \cup A^c = U$. Double: $(A^c)^c = A$. Absorption: $A \cup (A \cap B) = A$; $A \cap (A \cup B) = A$. De Morgan: $(A \cup B)^c = A^c \cap B^c$; $(A \cap B)^c = A^c \cup B^c$.

Compact examples

If $A = \{1,2,3\}$, $B = \{2,3,4\}$, $U = \{1,2,3,4,5\}$: $A \cap B = \{2,3\}$; $A \cup B = \{1,2,3,4\}$; $A - B = \{1\}$; $B - A = \{4\}$; $A^c = \{4,5\}$; $|P(A)| = 8$.

Lec 02 - Functions

Function: assigns exactly one output to each input. $f: A \rightarrow B$. If $a \in A$ maps to $b \in B$, $f(a) = b$. Domain = A. Co-domain = B. Range/Image = $\{f(a) \mid a \in A\}$; Range $\subseteq$ Co-domain.

Equal functions	Same domain, same co-domain, and $\forall x, f(x) = g(x)$
Injective / 1-1	Different inputs give different outputs; $f(a) = f(b) \Rightarrow a = b$
Many-to-one	$a_1 \neq a_2$ and $f(a_1) = f(a_2)$
Surjective / Onto	Every $b \in B$ has a preimage; Range = B
Bijjective	Injective and surjective
Inverse f^{-1}	If $f(a) = b$ then $f^{-1}(b) = a$; function inverse exists when f is bijective
Composition	If $g: A \rightarrow B$ and $f: B \rightarrow C$, $(f \circ g)(a) = f(g(a))$

Inverse algebra: $y = f(x)$, swap x and y, solve for y. Example: $f(x) = 2x + 3 \Rightarrow f^{-1}(x) = (x - 3)/2$.

Compact examples

$f(x) = 2x + 3$, $g(x) = 3x + 2$: $(f \circ g)(x) = f(3x + 2) = 6x + 7$; $(g \circ f)(x) = g(2x + 3) = 6x + 11$. If $A = \{a,b,c\}$, $B = \{1,2,3\}$, $f(a) = 1, f(b) = 2, f(c) = 3$: bijective.

Lec 03 - Sequences

Sequence: ordered list; order matters; repetition allowed. Terms may start $a_0, a_1, a_2, \dots$ or $a_1, a_2, \dots$

Arithmetic	$a, a+d, a+2d, \dots$; common difference d
nth term	$T_n = a + (n-1)d$
Sum	$S_n = n/2 [2a + (n-1)d] = n/2(a + T_n)$
Geometric	$a, ar, ar^2, \dots$; common ratio r
nth term	$T_n = ar^{n-1}$
Sum, $r \neq 1$	$S_n = a(1-r^n)/(1-r) = a(r^n-1)/(r-1)$
Σ	Σ from $k=m$ to n of $a_k = a_m + a_{m+1} + \dots + a_n$
Π	Π from $k=m$ to n of $a_k = a_m \times a_{m+1} \times \dots \times a_n$
Factorial	$n! = n(n-1)\dots 2 \times 1$; $0! = 1$

Standard sums: $\sum_{k=1}^n k = n(n+1)/2$; $\sum_{k=1}^n k^2 = n(n+1)(2n+1)/6$.

Compact examples

Arithmetic $a=3, d=4$: $T_{10} = 3 + 9(4) = 39$; $S_{10} = 10/2(3+39) = 210$. Geometric $a=5, r=2$: $T_6 = 5 \times 2^5 = 160$.

Lec 04 - Counting I

Sum rule	Mutually exclusive alternatives: $n_1 + n_2 + \dots + n_k$
Product rule	Sequential choices: $n_1 \times n_2 \times \dots \times n_k$
Subtraction	Add counts and subtract overlap
Division rule	If each outcome counted d times: distinct outcomes = n/d
Pigeonhole	N objects in k boxes $\Rightarrow$ at least one box has $\lceil N/k \rceil$ objects
Permutation no repetition	$P(n,r) = n!/(n-r)!$
Arrange all n	$P(n,n) = n!$
Permutation with repetition	n objects for r positions: n^r
Identical objects	$n!/(n_1!n_2!\dots n_k!)$

Restricted arrangements: use available choices for each position; without repetition, choices reduce after each selected position.

Compact examples

2 letters + 3 digits, repetition allowed: $26^2 \times 10^3$. Three digits from 1..8 without repetition: $8 \times 7 \times 6$. Odd three-digit number from 1..8 without repetition: last digit 4 choices, then $7 \times 6 \Rightarrow 168$. LEVEL arrangements: $5!/(2!2!)$.

Lec 05 - Counting II

Combination	Unordered selection; order does not matter
Without repetition	$C(n,r) = n!/[r!(n-r)!]$
Symmetry	$C(n,r) = C(n,n-r)$; $C(n,0) = C(n,n) = 1$
With repetition	$C(n+r-1,r) = (n+r-1)!/[r!(n-1)!]$
Binomial theorem	$(x+y)^n = \sum_{k=0}^n C(n,k)x^k y^{n-k}$
Coefficient	Coeff. of $x^n y^k$ in $(x+y)^n$ is $C(n,k)$
General coefficient	Coeff. of $x^n y^k$ in $(ax+by)^n$ is $C(n,k)a^n b^k$
Pascal identity	$C(n+1,k) = C(n,k-1) + C(n,k)$
Grid paths	(0,0) to (m,n): $C(m+n,m) = C(m+n,n)$; via point: multiply path counts

Pascal rows: $n=0$: 1; $n=1$: 1 1; $n=2$: 1 2 1; $n=3$: 1 3 3 1; $n=4$: 1 4 6 4 1; $n=5$: 1 5 10 10 5 1; $n=6$: 1 6 15 20 15 6 1; $n=7$: 1 7 21 35 35 21 7 1; $n=8$: 1 8 28 56 70 56 28 8 1.

Compact examples

$C(10,3) = 120$. Select 6 objects from 4 types with repetition: $C(4+6-1,6) = C(9,6)$. Coeff. of $x^{12}y^{13}$ in $(x+y)^{25}$ is $C(25,13) = C(25,12)$. Paths (0,0) to (5,3): $C(8,3) = 56$.

Lec 06 - Recursion

Recursive definition: defines a value using smaller/simpler values. Contains base case and recursive step. A recursive function $f(n)$ can define a sequence $a_0, a_1, a_2, \dots$ where $f(i) = a_i$.

Factorial	$F(0)=1; F(n)=nF(n-1); n! = n(n-1)\dots 1$
Sum 1..n	$S(0)=0; S(n)=S(n-1)+n; S(n)=n(n+1)/2$
Arithmetic	$a_1=a; a_n=a_{n-1}+d$
Geometric	$a_1=a; a_n=r a_{n-1}$
Fibonacci	$F_0=0, F_1=1; F_n=F_{n-1}+F_{n-2}$
Pascal	$C(n,0)=C(n,n)=1; C(n,k)=C(n-1,k-1)+C(n-1,k)$
Power	$p(0)=1; p(n)=a \cdot p(n-1)=a^n$
GCD	$\text{gcd}(a,b)$: if $a=0$ return b ; else $\text{gcd}(b \bmod a, a)$

Compact examples

If $L(0)=10$ and $L(n)=L(n-1)+5$: $L(1)=15, L(2)=20, L(3)=25$. Fibonacci: $0, 1, 1, 2, 3, 5, 8, 13$. $4! = 4 \times 3 \times 2 \times 1 = 24$. Power: $p(0)=1; p(n)=a \cdot p(n-1)$.

Lec 08 - Graph Theory

Graph: $G=(V,E)$, V =set of vertices/nodes, E =set of edges/connections. Edge endpoints are the vertices joined by the edge.

Directed graph	Edges have direction; (u,v) starts at u and ends at v
Undirected graph	Edges have no direction
Weighted graph	Edges have weights/values/costs
Simple graph	No loops; no multiple edges
Multigraph	Multiple edges may connect same vertices
Loop/self-edge	Edge from vertex to itself, e.g., (v,v)
Pseudograph	May contain loops and multiple edges

Graph terminology

Adjacent vertices: joined by an edge. Incident edge: edge touching a vertex. Degree $\deg(v)$: number of incident edges in undirected graph; a loop contributes 2. Directed: indegree $\deg^-(v)$ =incoming edges; outdegree $\deg^+(v)$ =outgoing edges.

Handshaking: $\sum \deg(v) = 2|E|$. Directed graphs: $\sum \deg^-(v) = \sum \deg^+(v) = |E|$. Maximum edges in simple graph with n vertices: $n(n-1)/2$.

Walk: sequence of vertices/edges. Path: walk between vertices. Simple path: no repeated vertices. Circuit: starts and ends at same vertex. Cycle/simple circuit: no repeated vertices except first=last. Connected: path between every pair of vertices. Subgraph $H=(W,F)$: $W \subseteq V$ and $F \subseteq E$. Hamiltonian path: visits every vertex once. Hamiltonian circuit: visits every vertex once and returns to start.

Adjacency matrix: rows/columns are vertices; entry 1 if edge exists, 0 otherwise. Undirected simple graph matrix is symmetric. Adjacency list: each vertex stores neighbours. BFS: queue. DFS: stack/recursion.

Compact examples

Simple graph with $n=8$: maximum edges = $8 \times 7/2 = 28$. If 15 vertices each degree 5: total degree 75 is odd, so impossible. Weighted path $A \rightarrow C \rightarrow D \rightarrow E$ with 2,3,7 has total 12.

Lec 09 - Trees

Tree: connected undirected graph with no simple circuits/cycles. Forest: graph with no simple circuits, consisting of disjoint trees. For a tree $G=(V,E)$: connected, acyclic, and $|E|=|V|-1$.

Rooted tree	One vertex designated root; edges directed away from root
Root	Node with no parent
Parent/Child	Directly above / directly below
Leaf	Node with no children
Siblings	Nodes with same parent
Degree	Number of children
Depth	Distance/path length from root to node
Height	Maximum depth
Ancestor	Node on path upward to root
Descendant	Node in subtree below

Binary tree: rooted tree where each node has at most 2 children. Full binary tree: every node has 0 or 2 children. Complete binary tree: all levels full except possibly last; last level filled left to right. Height of complete binary tree with n nodes: $\lceil \log_2 n \rceil$.

Heap: complete binary tree with heap property. Max-heap: parent value $\geq$ child values. Min-heap: parent value $\leq$ child values. 1-based heap array: root= $A[1]$; parent(i)= $\lfloor i/2 \rfloor$; left(i)= $2i$; right(i)= $2i+1$; max-heap $A[\text{parent}(i)] \geq A[i]$; min-heap $A[\text{parent}(i)] \leq A[i]$.

Spanning tree: subgraph of connected G that includes all vertices, has no cycles, and has $|V|-1$ edges. MST: spanning tree with minimum total edge weight. Prim: build tree incrementally using the lowest-cost edge connected to the current tree that does not form a cycle.

Compact examples

Tree with 15 vertices has 14 edges. Complete binary tree height: $n=7 \Rightarrow \lceil \log_2 7 \rceil = 3$; $n=15 \Rightarrow 4$; $n=16 \Rightarrow 4$. Heap array $A[1]$ root; node $i=5$ has parent $\lfloor 5/2 \rfloor = 2$, left child 10, right child 11.

Lec 10 - Big O Notation

Algorithm: step-by-step method to solve a problem. Time complexity: growth of running time. Space complexity: growth of memory use. Input size: n . Running time: count basic operations, comparisons, arithmetic, assignments.

Big O: upper bound of growth. $f(n)=O(g(n))$ if $\exists$ constants $c>0$ and $n_0>0$ such that $0 \leq f(n) \leq c \cdot g(n)$ for all $n \geq n_0$.

Growth order: $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n) < O(n!)$.

$O(1)$	Constant statements; assignment, arithmetic, comparison
$O(\log n)$	Input repeatedly halves/doubles
$O(n)$	One loop over n items
$O(n \log n)$	Common efficient divide/sort growth
$O(n^2)$	Two nested n loops; triangular nested loops
$O(2^n)$	Exponential branching
$O(n!)$	Factorial arrangements/brute force permutations

Simplification: keep dominant term; remove constants and lower-order terms. $5n+20=O(n)$; $3n^2+8n+10=O(n^2)$; $n+\log n+6=O(n)$; $2^n+n^3=O(2^n)$. Sequential code: add costs, dominant term remains. Nested code: multiply loop counts.

Common code patterns

Single statement: $O(1)$. for $i=1..n$: $O(n)$. for $i=1..n$, for $j=1..n$: $O(n^2)$. for $i=1..n$, for $j=1..i$: $O(n^2)$. while $n>1$, $n=n/2$: $O(\log n)$.